

# Hieu Pham

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## Profile

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I'm a career changer with 5+ years of experience in financial data analysis.

## Skills

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**Programming Languages** C, C++, Bash, PL/SQL, TypeScript, Go

**Development Tools** Git, Linux, Make, GDB, Docker, Docker Compose, Vim

## Projects

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### **Web Server** - HTTP Server from scratch

- Engineered a low-level **C++ HTTP server** via POSIX sockets, handling TCP connections and serving static HTML, CSS, Javascript, and image files
- Delivered full GET, POST, and DELETE support with spec-compliant status codes and response header
- Enabled concurrent browser and client support via poll-based non-blocking I/O without threads

### **Interception** - Dockerized Full Stack Web App

- Architected a multi-container web application using **Docker** and **Docker Compose**, running services in isolated, lightweight environments
- Configured **Nginx** as a reverse proxy with custom SSL/TLS encryption (HTTPS), managing secure traffic routing, port bindings, and optimized request handling for backend services
- Integrated a custom **WordPress** frontend connected to a dedicated **MariaDB** database with persistent volume management

### **Angry Cube** - A pseudo 3D game

- Implemented a 2D raycasting graphics engine from scratch in **C**
- Developed robust 2D linear vector mechanics for player movements and collision detection

### **SuperShell** - A lightweight shell interpreter

- Engineered a POSIX-compliant Unix shell in **C** with multi-process execution, pipeline routing and I/O redirection using system calls
- Designed a custom memory arena allocator to optimize heap usage and prevent memory fragmentation and leaks
- Built a custom lexer and a parser adhering to standard shell token recognition rules
- Implemented unit testing to ensure the robustness of the code base

### **Huff** - A text compression tool

- Implemented the Huffman algorithm to compress texts

### **Cfit** - An activity visualizer

- Explored a Developer Kit library for Garmin devices
- Analyzed the activity data and created a visualization using Raylib

### **Dining Philosophers** - Concurrency and Parallelism

- Implemented a multi-threaded and multi-process program to solve the classic dining philosopher problem using POSIX pthread and semaphore

**Professional Experience** \_\_\_\_\_

**Senior Data Analyst** (*Financial Recovery Technologies*) 2019-2024

- Exercised full accountability and project management skills by overseeing the lifecycle of a class action case
- Crafted **PL/SQL** scripts to automate calculation of clients’ entitled losses
- Translated highly technical legal and financial data into clear, actionable information for diverse clients

**Education** \_\_\_\_\_

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|-------------------------|-----------------------------------------------|
| Hive Helsinki           | Software Engineer                             |
| Suffolk University      | Master in Business Analytics                  |
| Northeastern University | Bachelor in Finance and Accounting Management |